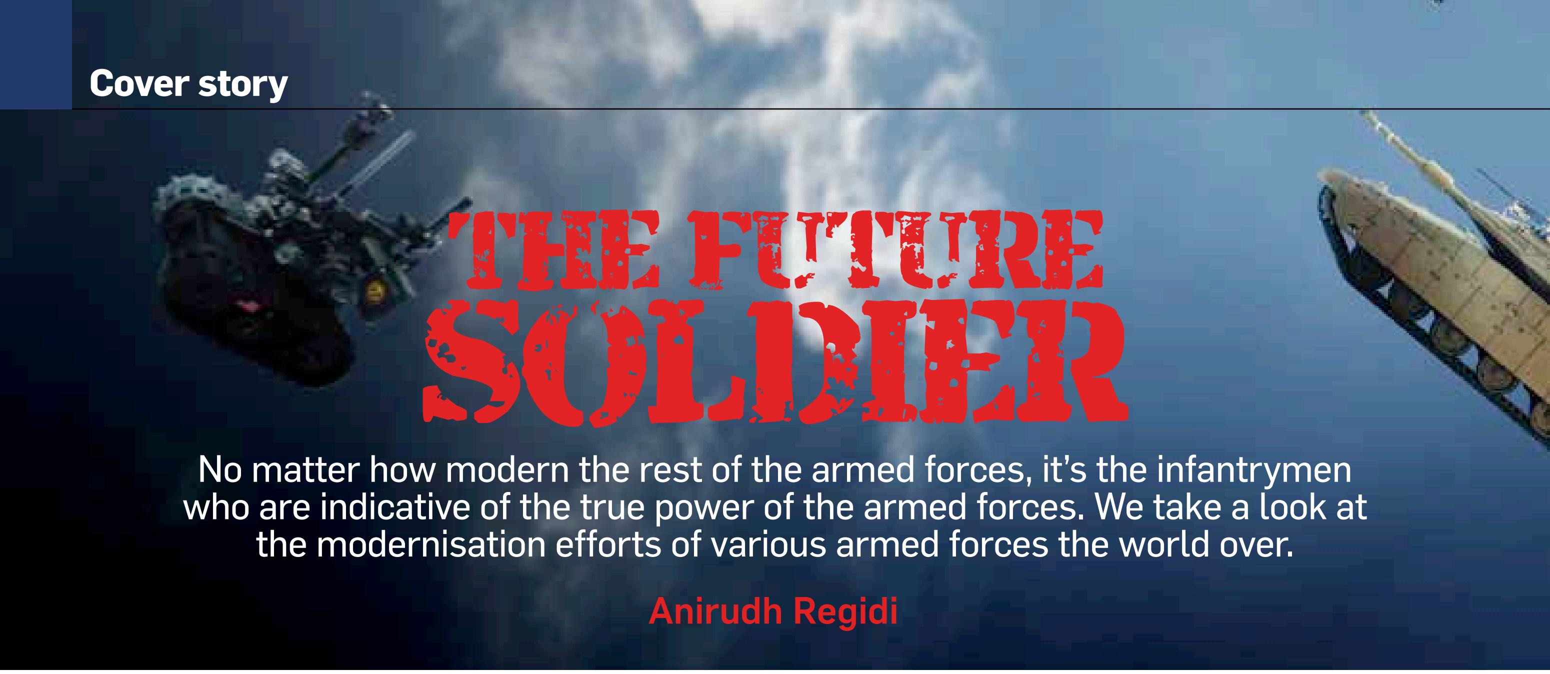

THE FUTURE SOLDIER

No matter how modern the rest of the armed forces, it's the infantrymen who are indicative of the true power of the armed forces. We take a look at the modernisation efforts of various armed forces the world over.

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The heart of every army is the infantrymen. Every military operation is designed around keeping the soldier alive and well, whether it be via the employment of drones, the CAS, etc. As long as the soldier still lives, an army can still fight. To achieve this goal, many armies the world over are working on various future soldier concepts including India's own F-INSAS program (Futuristic Infantry Soldier As a System). The goals of such programs are very simple, to increase the lethality, survivability, mobility, battle command, and training of their soldiers. The following aspects of the basic infantryman are all set to change in the near future.

A HIGHER CALIBER

The first, most basic step in increasing the lethality of a soldier is by giving him a weapon that is more effective than his enemy's. The primary combat weapon of any infantry soldier is his assault rifle. It comes in many shapes and forms, from bullpup layout of the L82A1 to the standard M-16 layout.

The bullets that these rifles fire are of various calibers and each have their distinct advantages and disadvantages. The standard NATO round is of the 5.56mm caliber, whereas the soviets and countries of the soviet bloc rely on the 7.62mm round. The NATO round is smaller and lighter and hence has lower stopping and penetrative power at medium ranges, but it's far more accurate and can travel farther and faster than the 7.62mm soviet rounds. On the other hand, the 7.62mm round offers amazing penetration and stopping power but doesn't have the range and the heavier round means that the recoil is significantly higher.

The problem is that the 5.56mm now is a bit inadequate for when faced with modern infantry armour and the 7.62mm is unnecessarily powerful and inaccurate. The answer is a compromise between the two calibers. Known as the 6.8 remington, the round actually has a caliber of 7mm but offers a good compromise between range, velocity, stopping power and accuracy. This will most probably be the round of the future.

TARGETING COMPUTERS

Sniping has always been a fine art and it takes rigorous training and mental conditioning to make a good sniper. A sniper needs to be taught to measure distance using environmental markers, adjust for wind shear, bullet drop and atmospheric pressure. In the middle of all this, the sniper needs to be aware of his breathing and has to time his shot perfectly. This is where the modern targeting computer comes in. The current spotter-sniper team is usually equipped with a ballistic computer that can give estimates as to where a bullet will drop based on various factors that can be entered into the computer.

OICW

The Objective Individual Combat Weapon or OICW was supposed to be to replace the M-16 that was the de-facto weapon for NATO soldiers at the time. The Weapon was supposed to incorporate an assault rifle, laser range finder, target designator, a shotgun, grenade launcher and an all-weather day and night scope. Unlike most such weapons, this wasn't one of those "too-good-to-be-true" dreams. The weapon actually worked as advertised. However, it proved to be too expensive to manufacture and too heavy to be comfortable. One can't have everything we suppose.

TrackingPoint, a startup based in Texas has come up with a sniper scope that makes all the adjustments for you. Think of it as a "smile-detection" shooting mode on cameras that take the picture only when everyone in the picture is smiling. The in-built computer and the in-built range finder will do all the calculations for you and can, if you wish it to, take the shot when the timing is just perfect.

Now this is just a startup we're talking about, but many armed forces the world over have been experimenting with similar technology for a while now and while they don't claim to do what TrackingPoint claims just yet, that's the goal they aim to achieve.

BATTLEFIELD NETWORKING

Another aim of the modernisation of the future soldier is the connected soldier. Information is key on the battlefield and when you don't have information on your own troops, you know that you're in a tight spot. The hope is that troops can be equipped with GPS sensors and various encrypted radios that will allow them to communicate with each other and with the commanders themselves.

In this age of the Internet we can't really imagine a world where Google or